

Rank- r Synchronization Defects and Projective Kummer Geometry

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Abstract

This note extends the two-generator CRT synchronization and projective Kummer-slope calculus to arbitrary rank r . For bases $a_1, \dots, a_r$, squarefree moduli $d = \prod_i q_i$, and local discrete-log rows

$$u_i = (u_{i1}, \dots, u_{ir}) \in (\mathbb{Z}/(q_i - 1)\mathbb{Z})^r,$$

global membership in $\langle a_1, \dots, a_r \rangle \bmod d$ is exactly membership of the target log vector in the image of a single exponent lattice map. The resulting synchronization defect quotient has an exact size formula. At a prime ℓ , the first-level defect is the cokernel of a Kummer log matrix over $\mathbb{F}_\ell$. For two primes the rank defect is equality of projective Kummer points in $\mathbb{P}^{r-1}(\mathbb{F}_\ell)$, outside star rows. Hyperplane events are controlled by one-radical Kummer fields, and the star atom is controlled by the full rank- r Kummer field. The proofs are finite algebra plus elementary Kummer independence by valuations.

1 Rank- r CRT synchronization

Let $a_1, \dots, a_r \in \mathbb{Z} \setminus \{0\}$. Let $d = q_1 \cdots q_s$ be squarefree with $(d, a_1 \cdots a_r) = 1$. Put

$$G_d = (\mathbb{Z}/d\mathbb{Z})^*, \quad \Gamma_d(\mathbf{a}) = \langle a_1, \dots, a_r \rangle \subset G_d.$$

For each i , choose a primitive root $g_i \bmod q_i$, and write

$$a_j \equiv g_i^{u_{ij}} \pmod{q_i}, \quad c \equiv g_i^{w_i} \pmod{q_i}.$$

Define the exponent-lattice map

$$\Phi_d : \mathbb{Z}^r \longrightarrow \bigoplus_{i=1}^s \mathbb{Z}/(q_i - 1)\mathbb{Z}, \quad (n_1, \dots, n_r) \longmapsto \left(\sum_{j=1}^r n_j u_{ij} \right)_{i=1}^s.$$

Theorem 1 (Rank- r squarefree lattice membership). *With notation above,*

$$c \in \Gamma_d(\mathbf{a})$$

if and only if

$$(w_1, \dots, w_s) \in \text{im}(\Phi_d).$$

Proof. By the Chinese remainder theorem,

$$G_d \simeq \prod_{i=1}^s \mathbb{F}_{q_i}^*.$$

For an exponent vector $n = (n_1, \dots, n_r)$, the image of

$$a_1^{n_1} \cdots a_r^{n_r}$$

in the i -th component is

$$g_i^{\sum_j n_j u_{ij}}.$$

Thus a single global exponent vector n sends $\mathbf{a}^n$ to c modulo every q_i exactly when

$$\sum_j n_j u_{ij} \equiv w_i \pmod{q_i - 1}$$

for all i . This is precisely the condition that the log vector $(w_i)_i$ belongs to the image of Φ_d . $\square$

Definition 1 (Rank- r synchronization defect). *Let*

$$\Gamma_{q_i}(\mathbf{a}) = \langle a_1, \dots, a_r \rangle \subset \mathbb{F}_{q_i}^*, \quad \Pi_d(\mathbf{a}) = \prod_{i=1}^s \Gamma_{q_i}(\mathbf{a}).$$

The rank- r synchronization defect quotient is

$$\mathcal{S}_d(\mathbf{a}) = \Pi_d(\mathbf{a}) / \Gamma_d(\mathbf{a}),$$

where $\Gamma_d(\mathbf{a})$ is embedded diagonally through CRT.

Proposition 1 (Exact subgroup-size formula). *For squarefree $d = \prod_i q_i$,*

$$|\Gamma_d(\mathbf{a})| = \frac{\prod_{i=1}^s |\Gamma_{q_i}(\mathbf{a})|}{|\mathcal{S}_d(\mathbf{a})|}.$$

Proof. Under CRT, $\Gamma_d(\mathbf{a})$ is a subgroup of

$$\Pi_d(\mathbf{a}) = \prod_i \Gamma_{q_i}(\mathbf{a}).$$

By definition,

$$|\mathcal{S}_d(\mathbf{a})| = [\Pi_d(\mathbf{a}) : \Gamma_d(\mathbf{a})].$$

Therefore

$$|\Pi_d(\mathbf{a})| = |\Gamma_d(\mathbf{a})| |\mathcal{S}_d(\mathbf{a})|.$$

Since $|\Pi_d(\mathbf{a})| = \prod_i |\Gamma_{q_i}(\mathbf{a})|$, the formula follows. $\square$

2 First-level ℓ -defect spaces

Let ℓ be prime and let T be a finite set of primes $q \equiv 1 \pmod{\ell}$, $q \nmid a_1 \cdots a_r$. For each $q \in T$, choose a primitive root $g_q \pmod{q}$ and define

$$x_\ell(q) = (u_{q1}, \dots, u_{qr}) \in \mathbb{F}_\ell^r, \quad a_j \equiv g_q^{u_{qj}} \pmod{q}.$$

Define the Kummer log matrix

$$M_\ell(T; \mathbf{a}) = \begin{pmatrix} x_\ell(q_1) \\ \vdots \\ x_\ell(q_s) \end{pmatrix} \in M_{s \times r}(\mathbb{F}_\ell).$$

Equivalently, define

$$\phi_{\ell, T} : \mathbb{F}_\ell^r \rightarrow \mathbb{F}_\ell^T, \quad n \mapsto (x_\ell(q) \cdot n)_{q \in T}.$$

Definition 2 (First-level ℓ -synchronization defect space). *The first-level ℓ -defect space is*

$$D_\ell(T; \mathbf{a}) = \text{coker}(\phi_{\ell,T}).$$

Its defect dimension is

$$\delta_\ell(T; \mathbf{a}) = \dim_{\mathbb{F}_\ell} D_\ell(T; \mathbf{a}) = |T| - \text{rank}_{\mathbb{F}_\ell} M_\ell(T; \mathbf{a}).$$

Theorem 2 (Rank formula for first-level ℓ -defects). *For every finite T ,*

$$\delta_\ell(T; \mathbf{a}) = |T| - \text{rank}_{\mathbb{F}_\ell} M_\ell(T; \mathbf{a}).$$

In particular, the first-level ℓ -synchronization obstruction is trivial if and only if the rows $x_\ell(q)$ span $\mathbb{F}_\ell^T$ as a map from exponent space, i.e. $\phi_{\ell,T}$ is surjective.

Proof. This is the rank-nullity theorem applied to the linear map

$$\phi_{\ell,T} : \mathbb{F}_\ell^r \rightarrow \mathbb{F}_\ell^T.$$

The cokernel has dimension $\dim \mathbb{F}_\ell^T - \text{rank}(\phi_{\ell,T}) = |T| - \text{rank} M_\ell(T; \mathbf{a})$. □

Corollary 1 (Defect lower bound). *If $|T| > r$, then*

$$\delta_\ell(T; \mathbf{a}) \geq |T| - r.$$

Proof. The matrix $M_\ell(T; \mathbf{a})$ has at most r columns, so its rank is at most r . Apply [Theorem 2](#). □

Remark 1. *This lower bound is not a pathology. It records the fact that one r -dimensional exponent vector cannot independently prescribe arbitrary first-level ℓ -data at more than r prime components. The analytic problem is to control which defect patterns are forced by Kummer distribution and which are random CRT overdetermination.*

3 Projective Kummer points

Definition 3 (Projective Kummer point). *For $q \equiv 1 \pmod{\ell}$, define the projective Kummer point*

$$\sigma_\ell(q; \mathbf{a}) = [u_{q1} : \cdots : u_{qr}] \in \mathbb{P}^{r-1}(\mathbb{F}_\ell)$$

when $x_\ell(q) \neq 0$. The star atom is the locus

$$x_\ell(q) = 0,$$

equivalently all a_j are ℓ -th powers modulo q .

Theorem 3 (Two-prime projective collision criterion). *Let $p, q \equiv 1 \pmod{\ell}$, and assume $x_\ell(p), x_\ell(q) \neq 0$. Then*

$$\text{rank} \begin{pmatrix} x_\ell(p) \\ x_\ell(q) \end{pmatrix} < 2$$

if and only if

$$\sigma_\ell(p; \mathbf{a}) = \sigma_\ell(q; \mathbf{a}) \quad \text{in } \mathbb{P}^{r-1}(\mathbb{F}_\ell).$$

Proof. Two nonzero vectors in $\mathbb{F}_\ell^r$ span a one-dimensional subspace if and only if they are scalar multiples. This is precisely equality of their projective classes. □

Corollary 2 (Two-generator determinant criterion recovered). *For $r = 2$, write*

$$x_\ell(p) = (u_p, v_p), \quad x_\ell(q) = (u_q, v_q).$$

Outside the star atom,

$$\sigma_\ell(p) = \sigma_\ell(q)$$

if and only if

$$u_p v_q - u_q v_p \equiv 0 \pmod{\ell}.$$

Proof. Two nonzero vectors in $\mathbb{F}_\ell^2$ are projectively equal if and only if the 2×2 determinant with those rows is zero. $\square$

4 Hyperplane atoms and Kummer fields

For $A = (A_1, \dots, A_r) \in \mathbb{F}_\ell^r$, define the rational element

$$\alpha_A = \prod_{j=1}^r a_j^{A_j} \in \mathbb{Q}^*$$

after choosing integer representatives $0 \leq A_j < \ell$.

Proposition 2 (Hyperplane atom criterion). *For $q \equiv 1 \pmod{\ell}$, $q \nmid a_1 \cdots a_r$,*

$$A \cdot x_\ell(q) = 0 \quad \text{in } \mathbb{F}_\ell$$

if and only if

$$\alpha_A \in (\mathbb{F}_q^*)^\ell.$$

Proof. Write $a_j \equiv g_q^{u_{qj}}$. Then

$$\alpha_A \equiv g_q^{\sum_j A_j u_{qj}} \pmod{q}.$$

Since $q \equiv 1 \pmod{\ell}$, an element $g_q^e \in \mathbb{F}_q^*$ is an ℓ -th power if and only if $e \equiv 0 \pmod{\ell}$. Thus α_A is an ℓ -th power modulo q exactly when

$$\sum_j A_j u_{qj} = A \cdot x_\ell(q) \equiv 0 \pmod{\ell}.$$

$\square$

Definition 4 (Projective hyperplane Kummer field). *For $A \neq 0$, the hyperplane atom $A \cdot x_\ell(q) = 0$ is controlled by*

$$K_{\ell,A} = \mathbb{Q}(\zeta_\ell, \alpha_A^{1/\ell}).$$

The star atom is controlled by

$$K_{\ell,\star} = \mathbb{Q}(\zeta_\ell, a_1^{1/\ell}, \dots, a_r^{1/\ell}).$$

Lemma 1 (Rational Kummer independence by valuations). *Let $a_1, \dots, a_r$ be distinct rational primes. If*

$$a_1^{e_1} \cdots a_r^{e_r} \in (\mathbb{Q}(\zeta_\ell)^*)^\ell, \quad 0 \leq e_j < \ell,$$

then

$$e_1 = \cdots = e_r = 0.$$

Proof. Let $\mathfrak{p}_j$ be any prime ideal of $\mathbb{Q}(\zeta_\ell)$ above the rational prime a_j . Taking the $\mathfrak{p}_j$ -adic valuation of the displayed identity gives

$$e_j \equiv 0 \pmod{\ell},$$

because the valuation of an ℓ -th power is divisible by ℓ , while the other rational primes a_k with $k \neq j$ have valuation 0 at $\mathfrak{p}_j$. Since $0 \leq e_j < \ell$, this forces $e_j = 0$. Applying this for every j proves the claim. $\square$

Corollary 3 (Rank- r star field degree). *For distinct rational primes $a_1, \dots, a_r$,*

$$[\mathbb{Q}(\zeta_\ell, a_1^{1/\ell}, \dots, a_r^{1/\ell}) : \mathbb{Q}(\zeta_\ell)] = \ell^r.$$

Proof. By Kummer theory over $\mathbb{Q}(\zeta_\ell)$, the degree equals the size of the $\mathbb{F}_\ell$ -span generated by the classes of $a_1, \dots, a_r$ in

$$\mathbb{Q}(\zeta_\ell)^* / (\mathbb{Q}(\zeta_\ell)^*)^\ell.$$

[Lemma 1](#) says these classes are linearly independent. Hence the span has dimension r , and the extension degree is ℓ^r . $\square$

Corollary 4 (Expected star density). *For distinct rational primes $a_1, \dots, a_r$, the star atom has relative Chebotarev density ℓ^{-r} among primes $q \equiv 1 \pmod{\ell}$, up to the usual exclusion of ramified primes.*

Proof. Over $\mathbb{Q}(\zeta_\ell)$, the star condition is complete splitting in the Kummer extension

$$K_{\ell, \star} / \mathbb{Q}(\zeta_\ell),$$

which has degree ℓ^r by [Corollary 3](#). Therefore the identity Frobenius class has relative density $1/\ell^r$. $\square$

5 Rank- r collision graphs and hypergraphs

Definition 5 (Rank- r Kummer slope collision graph). *For $Q > 0$, define $G_{\ell, r}(Q; \mathbf{a})$ to have vertices*

$$V_\ell(Q) = \{q \leq Q : q \equiv 1 \pmod{\ell}, q \nmid a_1 \cdots a_r, x_\ell(q) \neq 0\}.$$

Two vertices are adjacent if

$$\sigma_\ell(p; \mathbf{a}) = \sigma_\ell(q; \mathbf{a}).$$

Definition 6 (Rank-defect hypergraph). *For $k \geq 2$, define the k -uniform rank-defect hypergraph $H_{\ell, r}^{(k)}(Q; \mathbf{a})$ on the same vertex set by declaring*

$$\{q_1, \dots, q_k\}$$

to be a hyperedge when

$$\text{rank} \begin{pmatrix} x_\ell(q_1) \\ \vdots \\ x_\ell(q_k) \end{pmatrix} < \min(k, r).$$

Proposition 3 (Projective graph controls pairwise defects). *The edges of $G_{\ell, r}(Q; \mathbf{a})$ are exactly the two-vertex rank defects in $H_{\ell, r}^{(2)}(Q; \mathbf{a})$.*

Proof. This is [Theorem 3](#) applied to every pair. $\square$

Proposition 4 (Hyperplane certificate for rank defect). *If vectors $x_\ell(q_1), \dots, x_\ell(q_k)$ have rank $< k \leq r$, then there exist coefficients $B_1, \dots, B_k \in \mathbb{F}_\ell$, not all zero, such that*

$$\sum_{i=1}^k B_i x_\ell(q_i) = 0.$$

Equivalently, for every exponent vector $n \in \mathbb{F}_\ell^r$,

$$\sum_{i=1}^k B_i (x_\ell(q_i) \cdot n) = 0.$$

Proof. Rank $< k$ for the k row vectors is equivalent to linear dependence among those rows. That is precisely the existence of a nonzero coefficient vector $B = (B_i)$ satisfying the displayed relation. $\square$

6 Prime-power synchronization

The squarefree theory records synchronization across distinct prime components. Prime powers introduce vertical synchronization along the tower

$$q, q^2, \dots, q^e.$$

Let $q \nmid a_1 \cdots a_r$ be an odd prime. For $e \geq 1$, write

$$G_{q^e} = (\mathbb{Z}/q^e\mathbb{Z})^*, \quad \Gamma_{q^e}(\mathbf{a}) = \langle a_1, \dots, a_r \rangle \subset G_{q^e}.$$

Theorem 4 (Prime-power lifting exact sequence). *For $e \geq 1$, reduction modulo q^e induces an exact sequence*

$$1 \longrightarrow K_e \longrightarrow \Gamma_{q^{e+1}}(\mathbf{a}) \longrightarrow \Gamma_{q^e}(\mathbf{a}) \longrightarrow 1,$$

where

$$K_e = \Gamma_{q^{e+1}}(\mathbf{a}) \cap (1 + q^e\mathbb{Z}/q^{e+1}\mathbb{Z}).$$

Moreover

$$K_e \leq 1 + q^e\mathbb{Z}/q^{e+1}\mathbb{Z} \simeq (\mathbb{Z}/q\mathbb{Z}, +),$$

so

$$|K_e| \in \{1, q\}.$$

Proof. Reduction $G_{q^{e+1}} \rightarrow G_{q^e}$ sends each generator $a_j \bmod q^{e+1}$ to $a_j \bmod q^e$. Hence it maps $\Gamma_{q^{e+1}}(\mathbf{a})$ onto $\Gamma_{q^e}(\mathbf{a})$. Its kernel on the full unit group is

$$1 + q^e\mathbb{Z}/q^{e+1}\mathbb{Z},$$

which is cyclic of order q and canonically isomorphic to the additive group $\mathbb{Z}/q\mathbb{Z}$ by

$$1 + q^e t \longmapsto t \bmod q.$$

Intersecting this kernel with $\Gamma_{q^{e+1}}(\mathbf{a})$ gives K_e . Since the ambient kernel has prime order q , the subgroup K_e has order 1 or q . $\square$

Corollary 5 (Prime-power subgroup growth). *For every $e \geq 1$,*

$$|\Gamma_{q^{e+1}}(\mathbf{a})| = |\Gamma_{q^e}(\mathbf{a})| |K_e|,$$

with $|K_e| \in \{1, q\}$. Consequently

$$|\Gamma_{q^e}(\mathbf{a})| = |\Gamma_q(\mathbf{a})| q^{N_e},$$

where $0 \leq N_e \leq e - 1$ is the number of levels $1 \leq h < e$ for which K_h is nontrivial.

Proof. The size formula follows from the exact sequence in [Theorem 4](#). Iterating from level 1 to e gives the displayed expression. $\square$

Definition 7 (Vertical lifting defect). *The vertical lifting defect at level e is*

$$\nu_{q,e}(\mathbf{a}) = e - 1 - N_e.$$

It counts how many prime-power lifting levels fail to enlarge the subgroup by the full factor q .

Proposition 5 (General-modulus defect factorization). *Let*

$$d = \prod_{i=1}^s q_i^{e_i}$$

with $(d, a_1 \cdots a_r) = 1$. Then under CRT,

$$|\Gamma_d(\mathbf{a})| = \frac{\prod_i |\Gamma_{q_i^{e_i}}(\mathbf{a})|}{|\mathcal{S}_d^{\text{full}}(\mathbf{a})|},$$

where

$$\mathcal{S}_d^{\text{full}}(\mathbf{a}) = \left(\prod_i \Gamma_{q_i^{e_i}}(\mathbf{a}) \right) / \Gamma_d(\mathbf{a}).$$

Equivalently,

$$|\Gamma_d(\mathbf{a})| = \frac{\prod_i |\Gamma_{q_i}(\mathbf{a})| q_i^{N_{e_i}}}{|\mathcal{S}_d^{\text{full}}(\mathbf{a})|}.$$

Proof. The first formula is the same quotient-counting argument as [Proposition 1](#), applied to prime-power CRT components. The second formula substitutes [Corollary 5](#) for each component. $\square$

7 Insertion points for the EG203 stack

1. Paper 2 receives [Theorem 1](#), [Definition 1](#), and [Theorem 3](#) as a new final section: “Rank- r synchronization.”
2. The companion note on exact Γ -fiber local density receives [Theorem 4–Proposition 5](#) as the general-modulus extension.
3. The program overview receives the line:

The projective slope invariant extends from $\mathbb{P}^1(\mathbb{F}_\ell)$ to $\mathbb{P}^{r-1}(\mathbb{F}_\ell)$ for rank r .

4. The next computational script should implement $r = 3$, $\mathbf{a} = (2, 3, 5)$, by computing $x_\ell(q)$, the row-rank distribution of $M_\ell(T; \mathbf{a})$, and the spectral gap of $G_{\ell,3}(Q; \mathbf{a})$.